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Global Coreference Resolution via Local Resolution and Quotient-Graph Inference

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Abstract

This paper proposes that entity resolution does not require global mention-level reasoning which incurs a full quadratic attention compute cost. It may instead be performed through clusters formed over a quotient graph induced by locally resolved mention clusters. To accomplish this, a factorized architecture for global entity disambiguation or coreference resolution replaces the full document quadratic attention with a two-stage inference pipeline. The model separates semantic and contextual representations using independent frozen encoders and performs inference through local window-based clustering of mentions and cross-window matching over cluster-level representations constructed from a quotient graph. All pretrained encoders (RoBERTa and BGE) are kept frozen, and only lightweight scoring and aggregation modules are trained. Despite this constraint, the system achieves a CoNLL F_1 score of 0.8533 on CoNLL-2012 (gold mentions), comparable to fine-tuned end-to-end baselines. Empirically, we find that local window clustering captures most within-segment coreference structure, while cluster-level representations are sufficient for recovering long-range dependencies.

Keywords: coreference resolution, entity disambiguation, hierarchical clustering, frozen language models, window-based inference, quotient graph, representation factorization, efficient inference

1 Introduction

Coreference resolution aims to recover an equivalence relation over a set of nominal mentions $\mathcal{M} = \{m_1, \dots, m_n\}$ in a document, where $m_i \sim m_j$ indicates that two mentions refer to the same underlying entity. The induced partition $\mathcal{E} = \{E_1, \dots, E_k\}$ defines the entity structure of the document.

Modern coreference systems perform entity resolution through direct mention-level interaction over the entire document. Whether implemented through pairwise scoring, mention ranking, or transformer self-attention, the dominant paradigm assumes that global entity structure must be inferred from a space in which individual mentions remain the primary objects of reasoning throughout inference.

This paper investigates a different hypothesis: global entity resolution may not require global mention-level reasoning. Instead, most entity structure may be recoverable through local resolution within bounded contexts, after which global consistency can be reconstructed through interactions between locally induced entity clusters. If true, the problem can be reformulated from global mention inference to hierarchical entity construction. The proposed system consists of three components:

1. **Channel-separated representation:** Each mention is encoded using two frozen encoders that produce distinct semantic (surface-based) and contextual representations. These are kept separate throughout inference rather than collapsed into a single vector.

2. **Local window resolution:** The document is partitioned into fixed-length windows. Coreference links are first inferred independently within each window using a lightweight scoring model, producing local clusters via transitive closure.

3. **Cluster-level global matching:** Local clusters are treated as nodes in a quotient graph. Cross-window entity consistency is recovered by learning a matching function over cluster representations rather than directly comparing individual mentions.

This decomposition transforms global inference over M mentions into a two-level process: local inference over bounded windows and global reasoning over a reduced set of cluster nodes. This structure allows the system to avoid full-document attention while still recovering long-range entity structure through hierarchical composition. The design is motivated by empirical observations that scalar similarity between contextual embeddings loses substantial discriminative information compared to full vector interactions, and that local resolution within bounded contexts captures a large fraction of coreference structure. We further observe that cluster-level aggregation provides a more stable interface for cross-window linking than direct mention-level comparison in frozen encoder settings.

The main contributions of this work are:

- We formulate global coreference resolution as a hierarchical inference problem in which entity clusters, rather than mentions, become the primary objects of global reasoning.

- We introduce a quotient-graph formulation that lifts locally resolved mention clusters into a reduced entity space for cross-window inference.

- We demonstrate that strong full-document coreference performance can be achieved without full-document self-attention and without fine-tuning large pretrained encoders.

- We provide empirical evidence that local clustering captures sufficient entity structure to enable recovery of full-document entities through cluster-level reconciliation.

Unlike prior long-document coreference systems, the proposed approach does not approximate global mention interactions through sparse attention, sliding or overlapping windows, memory mechanisms, or document chunking followed by mention-level reconciliation. Instead, local resolution is treated as an explicit clustering stage whose outputs become the nodes of a quotient graph. Global inference is then performed exclusively over cluster representations. To our knowledge, existing neural coreference systems have not evaluated whether full-document entity structure can be recovered through this form of quotient-space reasoning.

2 Model Class Definition

We formalize the proposed system as a structured inference model over a partitioned mention space. The key property of the model class is the absence of any cross-partition mention-level interaction during local inference, followed by a single lifting step to a quotient space where global reasoning is performed exclusively over aggregated cluster representations. Let a document be represented as a sequence of tokens from which a set of mentions

$$\mathcal{M} = \{m_1, m_2, \dots, m_N\}$$

is extracted. The document is partitioned into a collection of disjoint, non-overlapping windows

$$\mathcal{W} = \{W_1, W_2, \dots, W_K\}$$

such that $W_i \cap W_j = \varnothing \forall i \neq j$ and

$$\bigcup_{i=1}^{K} W_i = \mathcal{M}$$

Each mention belongs to exactly one window via a deterministic assignment function $\pi : \mathcal{M} \to \mathcal{W}$. Let f_θ denote a frozen mention encoder producing representations:

$$\mathbf{h}_i = f_\theta(m_i) \in \mathbb{R}^d$$

For each window W_k, we define a restricted local model:

$$\mathcal{F}_k = \{f_\theta(m) \mid m \in W_k\}$$

Within each window W_k, we define a local coreference relation:

$$\sim_k \subseteq W_k \times W_k$$

computed via a scoring function:

$$s_\theta : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$$

Two mentions are linked if

$$m_i \sim_k m_j \iff s_\theta(\mathbf{h}_i, \mathbf{h}_j) > \tau$$

Let $\sim_k^*$ denote the transitive closure of this relation, inducing a partition $\mathcal{C}_k = W_k / \sim_k^*$. Stage A produces a collection of disjoint local clusters:

$$\mathcal{C} = \bigcup_{k=1}^{K} \mathcal{C}_k$$

Each cluster satisfies $C \subseteq W_k$ for exactly one k. We define a quotient mapping $q : \mathcal{M} \to \mathcal{C}$ inducing the quotient space $\mathcal{Q} = \mathcal{C}$. Each cluster $C \in \mathcal{Q}$ is treated as an atomic node in subsequent inference. Each cluster is embedded via a permutation-invariant function $E : \mathcal{Q} \to \mathbb{R}^d$.

$$E(C) = \mathrm{Agg}(\{\mathbf{h}_i \mid m_i \in C\})$$

We define a fully connected weighted graph $G_Q = (\mathcal{Q}, E_Q)$ over clusters with edge scoring function:

$$\sigma : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$$

Cluster-level coreference is defined as

$$C_a \sim_Q C_b \iff \sigma(E(C_a), E(C_b)) > \beta$$

Let $\sim_Q^*$ denote the transitive closure over cluster relations. The final entity partition is:

$$\mathcal{E} = \mathcal{Q} / \sim_Q^*$$

All cross-window reasoning is restricted to

$$\mathcal{Q} = \bigcup_k (W_k / \sim_k^*)$$

and not $\mathcal{M}$. Standard neural coreference models operate over a global mention interaction function

$$f : \mathcal{M} \times \mathcal{M} \to \mathbb{R}$$

possibly with pruning. In contrast, this model class enforces a strict factorization of inference into: (i) disjoint local equivalence construction and (ii) quotient-space graph inference over induced clusters.

3 Experimental Setup

We study coreference resolution over a set of mentions $\mathcal{M}$ extracted from documents in the CoNLL-2012 dataset. All experiments are conducted on the CoNLL-2012 English coreference benchmark (Pradhan et al., 2012). Crucially, all reported results and baseline comparisons assume gold mention boundaries. The system evaluated herein performs resolution exclusively over provided mentions and does not include a mention detection component. To ensure an exact and equitable comparison, all baseline results cited from prior literature correspond strictly to their respective gold-mention evaluation configurations.

- **Full-document evaluation:** Standard CoNLL evaluation over entire documents.
- **Window-constrained evaluation:** Evaluation restricted to fixed-length segments (used only for diagnostic analysis of Model A).
- **Intra-sentence evaluation:** Evaluation restricted to single sentences for ablation studies.

Model taxonomy We evaluate three model variants:

- **Model A (Local-only):** Windowed clustering without cross-window linking.
- **Model B (All-pairs):** Global mention-ranking over all mentions.
- **Model C (Hierarchical):** Model A followed by cluster-level cross-window matching.

4 Theoretical Foundation and Design Principles

We model coreference resolution as the construction of an entity partition over a set of mentions. Let $\mathcal{M} = \{m_1, \dots, m_n\}$ denote the set of mentions in a document. We define the coreference relation as $\sim \subseteq \mathcal{M} \times \mathcal{M}$ where $m_i \sim m_j$ if and only if both mentions refer to the same underlying entity. Since coreference is an equivalence relation, it induces a partition of $\mathcal{M}$ into disjoint entity sets $\mathcal{E} = \{E_1, \dots, E_k\}$ such that $E_a \cap E_b = \varnothing$ $(a \neq b)$ and

$$\bigcup_{a=1}^{k} E_a = \mathcal{M}$$

The objective of coreference resolution is to construct $\mathcal{E}$ from observed linguistic input. We formalize the structural assumptions underlying the proposed architecture.

Hypothesis 1 (Transitivity chain and locality) *Any two mentions of an entity can be connected through a chain of locally supported references. That is,* $\forall e \in \mathcal{E}, \forall u, v \in \mathcal{E}, \exists m_1, \dots, m_r \in \mathcal{E}$ *such that* $q(m_1) = q(u), q(m_r) = q(v)$ *and* $|i_{j+1} - i_j| \leq K$.

This is a direct consequence of the fact that for any entity that is recoverable globally must contain some non-zero signal of disambiguation in at least one window (otherwise, total signal in the entire corpus is 0 which renders the entity unrecoverable). This simple but crucial fact is what allows us to reconnect entities that are not directly clustered in any local windows.

Hypothesis 2 *Let* Z_s *and* Z_c *denote semantic and contextual features, and let* Y *denote the coreference indicator for a mention pair. Then neither signal is conditionally sufficient* $I(Y; Z_s \mid Z_c) > 0$ *and* $I(Y; Z_c \mid Z_s) > 0$.

Hypothesis 3 *There exists a finite-dimensional representation* $E(C) \in \mathbb{R}^d$ *that preserves sufficient information for entity-level comparison between clusters.*

The cluster representation preserves all information relevant to deciding whether two clusters belong to the same entity. There exists a representation function $E : \mathcal{C} \to \mathbb{R}^d$ such that

$$P(Y(C_a, C_b) | C_a, C_b) = P(Y(C_a, C_b) | E(C_a), E(C_b))$$

We encode each mention m using two frozen encoders:

$$g(m) = P_c(\hat{z}_{\text{context}}(m)) \parallel P_s(\hat{z}_{\text{semantic}}(m)), \tag{1}$$

where $\parallel$ denotes concatenation. Then any scalar similarity $\psi(g_i, g_j)$ is a many to one mapping. Let Y denote the coreference label for a pair of mentions and let $\psi(g_i, g_j) = \cos(g_i, g_j)$. By the Data Processing Inequality,

$$I(Y; \psi(g_i, g_j)) \leq I(Y; g_i, g_j)$$

This clearly establishes that a learned scorer operating on full channel vectors can exploit information unavailable to a scalar similarity measure.

4.1 Quotient Graph Construction

Let $G = (\mathcal{M}, \mathcal{R})$ be the mention graph. Stage A induces clusters $C_1, \dots, C_T$. We define the quotient graph $G/\sim_A$ whose nodes correspond to clusters. Each cluster is embedded as $E(C) \in \mathbb{R}^d$.

Proposition 1 *Let* $G = (M, R)$ *be the mention graph. Let Stage A induce equivalence classes* $C_1, \dots, C_t$. *Then any entity partition consistent with Stage A can be recovered by reasoning over* $G/\sim_A$ *rather than* G.

Proof Stage A induces an equivalence relation over local windows. Stage B adds cross-window links between clusters belonging to the same entity. The union of these relations forms an equivalence relation whose connected components coincide with $\mathcal{E}$. ∎

5 Intra-Sentence Baseline

The first test was scoped to a single sentence. Per token: a BGE embedding provides the semantic channel; a small learned bidirectional transformer (6.9M parameters) provides the contextual channel. Nominal heads are extracted via dependency parsing. For each mention k, the representation $r_k = [\mathbf{c}_k \,|\, \mathbf{s}_k]$ concatenates the contextual and semantic channel vectors. Per nominal pair (i, j)

$$\Phi(i,j) = \mathrm{MLP}\left([\, r_i \mid r_j \mid r_i - r_j \,]\right)$$

trained with binary cross-entropy; sigmoid-threshold decode into connected components via union-find. BGE-large + learned context reaches **CoNLL** $F_1 = 0.890$ across four corpora.

5.1 Semantic Channel Ablation

Variant	**CoNLL** F_1	**Error profile**
Semantic channel only (BGE-small)	0.772	High recall, precision 53%; over-links
Dual-stream framework	0.861	+0.089; precision 53% → 73%

Table 1: Semantic channel ablation (BGE-small) at sentence scale.

A single universal model with no context captures 90% of the intra-sentence signal. The contextual channel contributes almost entirely precision, suppressing false links that the semantic channel cannot distinguish.

5.2 Semantic Channel Scaling

Gains are concentrated in the hard semantic buckets where two mentions share no surface form. The bottleneck was semantic representation quality, not structural linking logic.

6 All-Pairs Global Resolution

The contextual channel in §5 was a small learned transformer trained from scratch. This section replaces it with a frozen RoBERTa-large backbone, training only the

Mention type	**BGE-small**	**BGE-large**	Δ
Pronouns	0.88	0.93	+0.05
Nouns, lexical variants	0.46	**0.66**	**+0.20**
Proper nouns, surface variants	0.50	**0.69**	**+0.19**
Proper nouns, identical surface	0.79	0.79	+0.00

Table 2: Link recall by mention type, BGE-small vs. BGE-large.

composition head. The goal is to establish how much of the disambiguation signal is already present in a general-purpose pretrained encoder before any task-specific adaptation.

6.1 Model

Neither large model is fine-tuned. The only trained component is a 2.6M-parameter head. For each gold mention i, the contextual channel supplies the RoBERTa token vector at the span start $\mathbf{c}_i^{(s)}$ and end $\mathbf{c}_i^{(e)}$. Let $\mathbf{s}_i$ denote the BGE vector of the mention's surface text. Each channel is projected independently:

$$g_i = [\, P_c([\mathbf{c}_i^{(s)} \mid \mathbf{c}_i^{(e)}]) \mid P_s(\mathbf{s}_i)\,] \in \mathbb{R}^{2048}$$

where $P_c \in \mathbb{R}^{1024\times 2048}$ and $P_s \in \mathbb{R}^{1024\times 1024}$ are learned, keeping the two information channels unblended within g_i. The channels are concatenated rather than summed: vector addition would collapse the channel decomposition and prevent the composition head from distinguishing semantic identity from contextual binding.

$$s(i,j) = \mathrm{MLP}_p\left([\, g_i \mid g_j \mid \mathbf{d}_{ij}\,]\right)$$

where $\mathbf{d}_{ij} \in \mathbb{R}^{32}$ is a learned mention-distance embedding. Mention-ranking marginal log-likelihood: each mention i softmaxes over $\{\varepsilon\} \cup \{j < i\}$, where ε is a learned null antecedent. The loss marginalises over all gold antecedents. Pairwise BCE over the same channels scored CoNLL $F_1 = 0.42$ (MUC 0.876, B^3 0.263, CEAFe 0.110) — a textbook over-merge. A global threshold trained on ~33% positive rate is miscalibrated at inference (~1%–2% positive rate), and one wrong high-confidence link chains two entities through transitive closure. Mention-ranking with a null antecedent fixes both. The 0.42 → 0.843 jump uses the same underlying vectors; it is entirely objective and decode.

6.2 Results

CoNLL $F_1 = 0.843$ (MUC 0.912, B^3 0.814, CEAFe 0.804). The intra-sentence restriction of the same system reaches 0.924. These are results with both backbones entirely frozen.

7 Local Intra-Window Resolution (Stage A)

This section analyzes the feasibility of intra-window coreference resolution under the proposed factorized architecture. The goal is not to evaluate an end-to-end system, but to estimate how much coreference structure can be recovered when all mentions are restricted to a bounded context window and encoded with frozen backbones. This setting removes cross-window dependencies by construction and therefore isolates the capacity of the local scoring model to recover entity structure within a constrained context. As such, the resulting metric is an **intra-window feasibility indicator** and is not comparable to full-document CoNLL-2012 scores or any SOTA systems. The document is partitioned into fixed non-overlapping windows of size $K = 256$ subtokens. Coreference links are computed independently within each window using the same frozen encoders and scoring architecture described in Section 1. Transitive closure is applied locally within each window to form clusters.

We report a single aggregated measure of intra-window clustering quality under this constraint: Intra-window resolution $F_1 = 0.8974$. This value represents the upper bound of local clustering performance under the frozen-encoder constraint. It should be interpreted strictly as a diagnostic measure of separability and not as a benchmark-comparable score. The high intra-window score indicates that most coreference structure is recoverable when contextual ambiguity is restricted to bounded segments. This supports the hypothesis that global reasoning in the full system is primarily a clustering and reconciliation problem rather than a representation learning problem. We emphasize again that this metric is used solely to validate the viability of Stage A as a local resolution module and is not used in any comparison against prior work.

8 Cross-Window Cluster Matching

Stage A resolves coreference within each fixed window independently. Its output is a set of per-window clusters: groups of mentions within the same window that refer to the same entity. These clusters are the nodes of the quotient graph $G/{\sim_A}$ that Stage A's union-find has already collapsed locally coreferent mentions into single nodes, reducing the problem from n mentions to $T \ll n$ clusters. Stage B's job is to connect the clusters in different windows that refer to the same entity. It does not re-examine individual mentions. It does not re-run Stage A. It operates purely on cluster-level representations.

Note that Stage B is not just a post-processing heuristic that stitches adjacent windows. Stage B is a learned scorer that compares every pair of clusters across every pair of windows; all $\binom{W}{2}$ window pairs, not just adjacent ones. An entity that appears in window 1 and window 8 with nothing in between is directly matched by Stage B without requiring a transitive chain through intermediate windows. This is the key difference from adjacent-only stitching and the reason all-pairs window comparison is necessary. Moreover, Stage B does not merge clusters greedily or arbitrarily. For each cluster C_a, Stage B scores all candidate matches C_b in other windows plus a learned null option. It links C_a to its best match only if the score exceeds the learned null bias; otherwise C_a remains unmerged. This is the same mention-ranking null-antecedent mechanism as Stage A, applied at the cluster level. The null bias is the precision control: setting it high produces fewer but more confident merges. Formally, Stage B connects edges of the quotient graph $G/\sim_A$ by solving, for each cluster C_a:

$$\hat{C}_b = \arg\max_{C_b \neq C_a} \sigma\left(E(C_a), E(C_b)\right)$$

merge C_a and $\hat{C}_b$ if and only if $\sigma > b$, where $b \in \mathbb{R}$ is the learned null bias and $E(\cdot)$ is the cluster encoder. Accepted merges are applied via union-find, and the final entity partition is the connected components of the resulting graph.

8.1 Cluster Encoder

Each Stage A cluster $C = \{m_1, \ldots, m_t\}$ is a set of mentions. Each mention m has a cached representation $\mathbf{m} = [P_c(\mathbf{c}_m) \mid P_s(\mathbf{s}_m)] \in \mathbb{R}^{2p}$ from Stage A. The cluster encoder $E(C)$ produces a single vector summarizing all members via attention pooling with a learned query $\mathbf{q} \in \mathbb{R}^{2p}$:

$$\begin{aligned} \alpha_m &= \frac{\exp(\mathbf{m}_m^\top \mathbf{q}/\sqrt{2p})}{\sum_{m'} \exp(\mathbf{m}_{m'}^\top \mathbf{q}/\sqrt{2p})} \\ E(C) &= \sum_m \alpha_m \mathbf{m}_m \in \mathbb{R}^{2p} \end{aligned}$$

This is the realization of Assumption 3: $E(C)$ is a learned sufficient statistic for the cluster's entity identity. The attention weights allow the encoder to focus on the most informative members (e.g., proper nouns over pronouns) rather than averaging uniformly. While simpler coordinate-wise algebraic operations (e.g., mean or max pooling) satisfy permutation invariance, they treat all underlying mention types uniformly. This risks washing out highly specific semantic anchors—such as proper nouns—with uninformative, recurring signatures from pronominal mentions. The learned query vector $\mathbf{q}$ in our attention pooling mechanism serves as a soft structural filter, dynamically weighting mentions that exhibit richer semantic footprints to preserve entity identity across distant windows.

8.2 Training

Stage B is trained on Stage A's predicted clusters, not on gold clusters split at window boundaries. This is deliberate: Stage B must learn to handle Stage A's errors, not assume perfect input. The gold label for a cluster pair (C_a, C_b) is positive iff the majority cluster-id of C_a's members equals the majority cluster-id of C_b's members in the full-document gold annotation. The loss is mention-ranking MLL over right-window candidates plus null, summed over all left clusters in all window pairs.

8.3 Results

	CoNLL	MUC	B^3	CEAFe
Stage A (window-split gold)	0.8974	0.9193	0.8932	0.8798
Stage A + B (full-doc gold)	**0.8533**	**0.9194**	**0.8315**	**0.8090**
Dobrovolskii (2021)	0.818	0.857	0.800	0.797
Caciularu et al. (2023)	0.836	0.874	0.820	0.813
Bohnet et al. (2023)	0.838	0.876	0.822	0.815
Luo et al. (2025)	0.851	0.889	0.835	0.829

Table 3: Full-document CoNLL-2012 test results on **gold mention boundaries** (Pradhan et al., 2012). All baseline configurations are mapped strictly to their gold-mention settings to isolate coreference logic from mention detection errors.

Cluster type	CoNLL	MUC	B^3	CEAFe
PROPN-only	0.839	0.840	0.838	0.839
PRON+NOUN	0.770	0.763	0.749	0.797
NOUN-only	0.750	0.753	0.750	0.748
PRON+PROPN	0.728	0.745	0.694	0.743
all-mixed	0.716	0.773	0.673	0.703
NOUN+PROPN	0.713	0.719	0.704	0.715
PRON-only	0.664	0.705	0.649	0.637

Table 4: Stage A + B chronological type breakdown, CoNLL-2012 test.

Evaluated directly against full-document gold, Stage A suffers a substantial loss due to entity fragmentation at window boundaries. Stage B exists specifically to recover these missing global links. It is therefore not a post-processing heuristic but the mechanism that reconstructs the full-document entity partition from locally resolved components. Stage B recovers ≈0.06 CoNLL points over Stage A alone on full-document gold, bringing the two-stage system to 0.8533. PRON-only (0.664) remains the hardest bucket: pronouns carry an uninformative signature inside the semantic channel, forcing Stage B to rely entirely on context profiles which naturally decay across window gaps.

9 Complexity Analysis

We analyze the computational complexity of the inference pipeline in terms of document length N, number of mentions M, window size K, and number of Stage A clusters C. The system consists of three components:

(i) windowed encoder processing

(ii) intra-window mention linking

(iii) cross-window cluster matching

The document is partitioned into $W = N/K$ windows. Each window is processed independently by a transformer encoder with self-attention cost $O(K^2)$. The total encoding cost is:

$$\begin{aligned} T_{\text{encode}} &= O\left(\frac{N}{K} \cdot K^2\right) \\ &= O(NK) \end{aligned}$$

This replaces full-document self-attention cost $O(N^2)$ with windowed attention over fixed-length segments. Let m_w be the number of mentions in window w. Stage A performs pairwise scoring within each window:

$$T_{\text{local}} = \sum_{w=1}^{W} O(m_w^2)$$

Assuming approximately uniform mention distribution $m_w \approx M/W$, we obtain:

$$\begin{aligned} T_{\text{local}} &= O\left(W \cdot (M/W)^2\right) \\ &= O(M^2/W) \end{aligned}$$

Substituting $W = N/K$ and $M = \rho N$,

$$T_{\text{local}} = O(\rho^2 NK)$$

Thus, Stage A scales linearly in N for fixed window size K and mention density ρ, and quadratically in ρ. Stage A produces C clusters across the document. Stage B evaluates pairwise interactions between clusters across windows:

$$T_{\text{cross}} = O(C^2)$$

Let $\bar{s} = M/C$ denote the average cluster size. Then $C = M/\bar{s}$, giving:

$$\begin{aligned} T_{\text{cross}} &= O\left(\frac{M^2}{\bar{s}^2}\right) \\ &= O\left(\frac{\rho^2 N^2}{\bar{s}^2}\right) \end{aligned}$$

Combining all components

$$T(N) = O(NK) + O(\rho^2 NK) + O\left(\frac{\rho^2 N^2}{\bar{s}^2}\right)$$

or equivalently

$$T(N) = O\left(NK(1+\rho^2)\right) + O\left(\frac{\rho^2 N^2}{\bar{s}^2}\right)$$

The proposed solution does not alter the fundamental quadratic upper bound of long-document coreference resolution; asymptotically, $T_{\text{cross}} = O(\rho^2 N^2/\bar{s}^2)$ remains quadratic with respect to token length N under a constant mention density ρ and bounded mean cluster size $\bar{s}$. However, the framework radically compresses the empirical operational footprint by transforming the primitive interaction space. Standard architectures execute $O(N^2)$ token-to-token or mention-to-mention interactions. By contrast, Stage B isolates quadratic interactions strictly to the quotient space nodes C, where $C \ll N$. Empirically across the CoNLL-2012 corpus, the average cluster size $\bar{s}$ acts as a powerful constant scaling down the coefficient of the quadratic term. Thus, while the asymptotic worst-case bounds are preserved, the absolute computational runtime is dramatically lower, transforming cross-window global matching into an efficient operation on a highly sparse entity topology.

10 Limitations and Future Work

Every result in this paper uses both encoders frozen, so the contextual channel is whatever RoBERTa supplies without any coreference adaptation, and RoBERTa was never trained to bind referents. The hardest buckets reflect exactly this: PRON-only clusters fall to 0.664 after Stage B because pronouns carry no semantic signature and the contextual channel alone must carry the decision. The decomposition isolates this limitation to a single, replaceable component.

Every result in this paper assumes gold mention boundaries: the system is evaluated as a linker given mentions. Extending the decomposition to predicted mentions is left to future work.

11 Conclusion

We presented a factorized architecture for global entity disambiguation that separates representation learning into frozen semantic and contextual channels and performs inference through a two-stage hierarchy consisting of local window clustering followed by cross-window cluster matching. Across all experiments, both encoders are kept frozen and only lightweight scoring and aggregation modules are trained. The full system achieves a CoNLL F_1 of 0.8533 on CoNLL-2012 under standard full-document evaluation. A non-windowed all-pairs variant reaches 0.843, indicating that the hierarchical decomposition preserves overall effectiveness while restructuring inference. The intra-window evaluation is used strictly as a diagnostic of local separability under bounded context and is not treated as a comparable benchmark metric. It serves to characterize the behavior of Stage A in isolation prior to global reconciliation. Empirically, results across intra-sentence, all-pairs, and hierarchical settings suggest that entity structure can be recovered using factorized representations combined with cluster-level matching, without fine-tuning large encoders or using full-document self-attention. Overall, the findings support the feasibility of decomposing entity disambiguation into local resolution followed by hierarchical merging over compressed cluster representations under frozen-encoder constraints.

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